# Vision-Based Gesture-to-Speech Translation for Sign Language

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Over 5% of the world's population suffers from a disabling hearing loss. As society rapidly becomes more interconnected, people with disabilities are usually left behind when dealing with new technologies: reducing this gap is therefore essential. Inclusive tools are a necessity for solving these challenges and deserve greater attention.

One active research branch in this area involves the implementation of computer-vision applications for gesture-to-speech translation. Training a Neural Network (NN) on automatic translation of sign language sentences (videos) is quite complex and requires a lot of computational power. Given our limited resources[1], we decided to implement our custom lightweight NN on a reduced dataset containing only alphabet spelling and numerical gestures.

The application will involve the training of a Convolutional Neural Network (CNN) on pre-existing dataset for Italian or American Sign Language[2]. Some letters are not included because they require a dynamic hand gesture that cannot be expressed in a static image. Previous scientific studies[3] that use similar approaches are going to be consulted as a baseline for the realization of our project.

Ideally, an application will be created that uses the NN found to automatically translate signs from a real-time video. To study the performance of our model, we aim at testing different hyperparameters and compare our results with some well-known architectures such as *ResNET50*, *MobileNetV2*, etc.

[1] MacBook Pro M3 8-core with 16GB RAM.

[2] The dataset we will use are the following (one or both depending on if we want to give double language option):

- American Sign Language Dataset: https://www.kaggle.com/datasets/grassknoted/asl-alphabet
- Italian Sign Language Dataset: https://www.kaggle.com/datasets/nicholasnisopoli/lisdataset

[3] Research papers used as sources:

- Pranjal Gogoi, B. Karsh, R.K. Karsh, R.H. Laskar, M.K. Bhuyan, (2025) "Vision-Based Real-Time Gesture-to-Speech Translation for Sign Language" in *Procedia Computer Science*
- Walizad, Mohammad Elham, and Mehreen Hurroo. (2020) "Sign language recognition system using convolutional neural network and computer vision." in *Int. J. Eng. Res. Technol*